Afeka College of Engineering, Tel-Aviv

Smart Thermal System for Patient Safety Monitoring

# Physical-Contact (Touch) Detection

Detailed Investigation: Methods, Reasoning, and Results

Waveshare 26984 80×62 px 3 synchronised views

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# 1 Problem, data, and evaluation

## 1.1 Task

The touch module detects *physical contact between patients* (covering the "physical contact" and "violent/sexual behaviour" scenarios) from three synchronised MLX-class thermal cameras — here the Waveshare 26984 at 80×62 px, 8 Hz. No optical imagery is used, preserving privacy. A frame is positive if two or more people are in physical contact in the scene.

## 1.2 Labels and the central constraint

Contact is annotated per frame in `contact_labels.csv`. The decisive fact for everything below is its scarcity:

- **188 positive frames** in the entire dataset, concentrated in only **four scenes**: `2ppl_fight` (50), `3pp_surprise` (66), `3pplhedroncolider` (53), `2ppl_hug` (19).
- All other scenes are contact-negative and act as **hard negatives**: single people, and crucially people who are *near but not touching* (`2ppl_walk`, `3ppl_dance`, `the_more_the_merrier`).

## 1.3 Evaluation protocol

Every method is scored on a common per-session *timeline* split (first 60% of each scene → train, next 15% → validation, last 25% → test), giving a fixed **test set of 695 frames (41 positive, 654 negative)**. Timeline (not random) splitting preserves temporal contiguity, required by the temporal models. Metrics: accuracy, precision, recall, F1, and false-alarm rate (FP/(FP+TN)). **F1** is primary; **recall** is the safety priority and **false-alarm rate** guards against alert fatigue (itself a safety risk in a ward). §5 revisits this single split with cross-validation.

## 1.4 Result at a glance

Table 1 previews the full arc; each row is detailed in its section.

Table 1: Progression of touch detectors on the common 695-frame test split.

| **Method** | **Core idea** | **F1** |
|---|---|---|
| Geometric multi-view (§2.1) | homography → floor distance | 26.8% |
| MV-STGCN (§2.2) | graph net on projected tracks | 25.2% |
| Thermo-X3D (§2.3) | 3-D CNN on raw volume | 41.7% |
| Image-plane voting V3 (§3.1) | box-gap + camera vote | 28.6% |
| + box-merge / router V4 (§3.4) | cleaner boxes; people routing | 35.5% |
| + blob-merge V6.3 (§3.5) | two bodies = one warm blob | 44.6% |
| + temporal V8 (§3.7) | morphological smoothing | 53.8% |
| + two-body merge V9 (§3.8) | reject 3-body clusters | 55.0% |
| + raw detector (config-D, §4) | raw for detection, residual for blobs | **60.8%** |
| *Honest cross-validated estimate of config-D (§5)* | | ∼44% |

# 2 Part A — The three designed detectors

The system design specified three contact detectors spanning a spectrum from classical geometry to deep appearance models. We implemented and evaluated all three first.

### 2.1 Geometric multi-view fusion

**Method.** For each camera, MobileNet-SSD detects people; each box is reduced to its *foot-point* (bottom-centre ≈ floor contact). A per-camera homography $H_k$ projects foot-points onto a common floor plane. Projections are clustered and cross-camera-validated into unique *actor positions*; any pair within a distance $\delta$ is flagged as contact.

**The homography problem.** Standard calibration needs heated targets at *known* floor coordinates, which were never surveyed. We therefore *self-calibrated*: in the calibration recording one person is visible to all three cameras, so that person's foot-point in each view is an image of one common floor point. Across the ∼146-frame walk, this yields image-to-image correspondences from which $H_{1\to 0}, H_{2\to 0}$ are solved by DLT inside a RANSAC loop (camera 0 as the reference plane). Distances are then in camera-0 floor-pixels, not metres, so $\delta$ is tuned empirically.

**Result.** F1 **26.8%** (precision 26.8%, recall 26.8%, FAR 4.6%).

| | **Pred: Contact** | **Pred: None** |
|---|---|---|
| **Truth: Contact** | **TP** = 11 | **FN** = 30 |
| **Truth: None** | **FP** = 30 | **TN** = 624 |

**Why it underperforms.** A diagnostic on true-contact frames showed that, with strict two-camera validation, the mean number of *surviving* actors was only **0.16** — the self-calibrated homography is accurate enough to localise but too noisy to fuse: the same person projects to different floor locations in different views, so cross-camera agreement (within the clustering radius) almost never holds. Relaxing the validation produces phantom duplicate actors instead. Two failure modes emerge that recur throughout: **(#1) homography noise** and **(#2) blob-merge** — when two people touch, they fuse into one detection, so there is no second box to measure.

### 2.2 MV-STGCN (multi-view spatiotemporal graph CNN)

**Method.** The same SSD+homography front-end produces actor nodes; a Kalman tracker (constant-velocity, correcting the ∼50 ms inter-camera I$^2$C phase shift) adds velocities. A sliding window of 16 actor-graphs feeds an adaptive-adjacency ST-GCN that classifies contact/no-contact. (Because our labels are binary with no actor positions, we injected SSD+fusion actor positions into the training events so the graph had node features.)

**Result.** F1 **25.2%** (precision 14.8%, recall **82.9%**, FAR **29.8%**).

| | **Pred: Contact** | **Pred: None** |
|---|---|---|
| **Truth: Contact** | **TP** = 34 | **FN** = 7 |
| **Truth: None** | **FP** = 195 | **TN** = 459 |

**Reading.** High recall but a 30% false-alarm rate: it fires on essentially every multi-person non-contact scene (`1person cig`, `2ppl_walk`, `3ppl_dance`). It inherits failure mode #1 directly — mis-projected duplicate actors land close together and look like contact.

### 2.3 Thermo-X3D (volumetric appearance network)

**Method.** A small (2+1)D factorised X3D encoder processes each camera's raw 16-frame thermal *volume*, with CBAM thermal attention to focus on hot voxels; the three per-camera features are late-fused and classified. It uses **no bounding boxes and no homography** — the designed failsafe for when contact merges two people into one blob.

**Result.** F1 **41.7%** (precision 48.4%, recall 36.6%, FAR 2.4%) — the best of the three designed detectors.

| | **Pred: Contact** | **Pred: None** |
|---|---|---|
| **Truth: Contact** | **TP** = 15 | **FN** = 26 |
| **Truth: None** | **FP** = 16 | **TN** = 638 |

**Reading.** Being detector- and homography-independent, it avoids both failure modes and is far cleaner (FAR 2.4%). Its misses are on `2ppl_fight` (0% recall there) — a *data* problem (only ∼90 positive training windows), not a method flaw. This result motivated the rest of the work: *a pixel/appearance signal beats the geometric pipeline.*

**Part A takeaway.** Homography is the weak link on this data; the best detector ignores it. We then asked whether a *simpler, homography-free* detector could do better.

# 3 Part B — A homography-free image-plane family

The idea: run SSD per camera, decide "touch" from the 2-D boxes in each image, and combine the three views. No homography, no world projection.

## 3.1 Image-plane voting (V1/V2/V3)

**Method.** A camera "votes touch" if two person-boxes have an edge-to-edge gap below $\tau$ pixels (0 = overlapping). Three ways to combine the three votes: **V1 majority** (≥2 cameras), **V2 unanimous** (all 3), **V3 relaxed** (≥2 vote, and the third corroborates — near-touch or a single merged blob — otherwise a third camera clearly seeing two separated people vetoes). $\tau$ tuned on validation.

**Result.**

| | **Prec** | **Rec** | **F1** | **FAR** |
|---|---|---|---|---|
| V1 majority | 18.5% | 56.1% | 27.9% | 15.4% |
| V2 unanimous | 14.7% | 53.7% | 23.0% | 19.6% |
| **V3 relaxed** | 19.2% | 56.1% | **28.6%** | 14.8% |

**Reading.** V3 is best. Counter-intuitively V2 (unanimous) is *worst*: requiring strict 3-camera agreement is so rare that tuning loosens $\tau$ to compensate, which floods false positives. Already comparable to the geometric detector, with no homography.

## 3.2 Error analysis of V3

We dissected V3's 18 missed contacts (FN) and 97 false alarms (FP):

- **FN cause 1 — blob-merge** (11 frames, all `2ppl_fight`): the fighters fuse into one box in 2 of 3 cameras (only one camera still sees two), so V3 gets one TOUCH vote and needs two.
- **FN cause 2 — threshold strictness** (7 frames, `3pp_surprise`): boxes are 1–6 px apart, just above $\tau$.
- **FP cause 1 — crowd proximity** (∼86 FP: `3pplhedroncolider`, `3ppl_dance`, `the_more_the_merrier`): three clustered people whose boxes overlap in every view without contact.
- **FP cause 2 — detector over-segmentation** (∼7 FP, `1person cig`): one person (or person + cigarette) split into two abutting boxes.

These four causes drove every subsequent improvement.

### 3.3 Targeted improvements V3.1–V3.5

Each idea was implemented and evaluated *separately* (re-tuned on validation):

| | Idea | Rec | F1 | FAR |
|---|---|---|---|---|
| V3.0 | baseline | 56.1% | 28.6% | 14.8% |
| V3.1 | merge-aware quorum | **92.7%** | 28.3% | 29.1% |
| V3.2 | temporal persistence ($\geq$2/3) | 43.9% | 25.9% | 12.2% |
| V3.3 | size-normalised gap | 75.6% | 30.2% | 20.3% |
| **V3.4** | merge over-segmented boxes | 58.5% | **32.0%** | 13.0% |
| V3.5 | OR Thermo-X3D ensemble | 36.6% | 42.3% | 2.3% |

**Reading.** **V3.4 (box-merge)** is the only free win: merging over-segmented boxes raised both precision and recall (F1 +3.4) by removing single-person false alarms. **V3.1 (merge-aware)** — counting a single merged blob toward the quorum — nearly eliminates the fight misses (recall 92.7%) but at a heavy precision cost. **V3.2 (persistence)** hurt (contacts are short). **V3.5** degenerated to Thermo-X3D alone: naively OR-ing the box detector with X3D adds only false positives, so tuning zeroed out the box branch.

### 3.4 Compound V4 / V5: merge-aware + box-merge + people-count routing

**Method.** Combine the winners: clean boxes (V3.4), merge-aware quorum (V3.1), and route by crowd size — trust the box logic on $\leq$2-person frames (where it catches fights and X3D fails), defer to Thermo-X3D on $\geq$3-person frames (where the box logic sprays). V5 is the same routing expressed as an explicit people-count switch.

**Result.** F1 **35.5%** (precision 24.3%, recall 65.9%, FAR 12.8%); V5 is numerically identical to V4. The crowd-veto eliminated `3ppl_dance` false alarms, and merge-awareness recovered `2ppl_fight` (84% recall — the scene X3D misses entirely). Remaining false alarms shifted to single/pair scenes where SSD detects a hot *object* (cigarette, heater) as a second "person" — which pure box geometry cannot distinguish from two people.

### 3.5 V6 — putting the thermal signal back in

The box family ignored temperature; both its remaining error types are thermal questions. We tested four thermal cues separately:

| | Idea | Rec | F1 | FAR |
|---|---|---|---|---|
| V6.1 | heat-bridge (warm path between boxes) | 36.6% | 39.0% | 3.2% |
| V6.2 | temperature box-rejection (drop hot objects) | 65.9% | 41.2% | 9.6% |
| **V6.3** | **blob-merge (two people in one warm component)** | 61.0% | **44.6%** | 7.0% |
| V6.4 | learned logistic combiner | 41.5% | 35.1% | 6.0% |

**Method & reading.** **V6.3** is the breakthrough and the first homography-free detector to *beat* Thermo-X3D. It segments warm connected components on the background-subtracted residual and flags contact when two detected people fall in the *same* component (their bodies have fused). This single idea fixes both error types at once: a cigarette is a *separate* component from its smoker (object false alarms collapse, `1person cig` 23→4 FP), and people merely standing close are separated by cool air (`3ppl_dance` →0 FP). **V6.1 (heat-bridge)** sampling temperature along the line between two boxes gave the best precision/FAR but cut recall (too strict). **V6.4**

**(learned combiner)** *underperformed* the hand rule — the first sign that learning loses on this little data.

### 3.6 V7 — blob-merge on crowds

**Method.** V6.3 deferred 3-person crowds to X3D, which fails `3pplhedroncolider`. V7 applies the blob-merge test to all regimes.

**Result.** F1 **44.6%** (V7b, recall 65.9%). It recovered `3pplhedroncolider` recall from 0% to 86%, but that scene's *negative* frames also trip the test (three colliding bodies are one blob), so the gained true positives were offset by new false positives — F1 unchanged. V7b is preferred over V6.3 at equal F1 because it covers all four contact scenes, including collisions.

### 3.7 V8 — temporal processing

**Method.** Contacts persist over time, so the per-frame errors are largely temporal noise. We applied, on the per-scene prediction stream: **(T1) morphological open+close** (remove short positive runs = flicker FPs; fill short negative gaps = dropout FNs); **(T2) centred window vote**; **(T3) a temporal logistic regression** on windowed features.

| | **Rec** | **F1** | **FAR** |
|---|---|---|---|
| **T1 morphology open+close** | 68.3% | **53.8%** | 5.4% |
| T2 centred window vote | 68.3% | 51.4% | 6.1% |
| T3 temporal logistic regression | 36.6% | 34.9% | 4.6% |

**Reading. T1** lifted F1 by **+9 points** (44.6→53.8), improving precision *and* recall. T3 (learned) again lost — the second confirmation that learned additions overfit here.

### 3.8 V9 — two-body merge + temporal

**Method.** The remaining gap was `3pplhedroncolider`'s sustained false positives. V9 adds a spatial discriminator: on crowds, fire only when a warm component contains *exactly two* person-centres (a genuine pairwise merge), rejecting the three-body cluster; then T1 temporal smoothing.

**Result.** F1 **55.0%** (precision 56.4%, recall 53.7%, FAR 2.6%). `3pplhedroncolider` false alarms dropped 19→4. A confidence-aware variant trades to recall 73.2% at F1 54.1%.

## 4 Part C — Preprocessing ablation (key finding)

**Question.** The Tateno preprocessing (Gaussian smooth → background subtract → residual) feeds the pipeline in two places: the SSD detector and the blob segmentation. Does it help? We retrained the SSD on *raw* frames and tested all combinations with the V9 detector (no X3D):

| **Config** | **Detector / blob input** | **F1** |
|---|---|---|
| A | Tateno SSD / residual blob | 54.5% |
| B | Tateno SSD / raw-temp blob | 8.7% |
| C | raw SSD / raw-temp blob | 0.0% |
| **D** | **raw SSD / residual blob** | **60.8%** |

**Finding — preprocessing should differ by task.**

- The blob test *needs* the residual (B, C collapse to ∼0): on raw temperature, bodies barely exceed the warm background, so connected-component segmentation cannot separate people.

- The residual *hurts* the detector (D 60.8% > A 54.5%): subtracting the background removes body signature the detector uses; the raw-trained SSD even reached a lower training loss.

**config-D** — raw SSD + residual blob + two-body merge + temporal — is the best detector at **F1 60.8%** (precision 50.8%, recall 75.6%, FAR 4.6%) on this split, with no homography and no contact training labels.

# 5 Part D — Reaching 65%? The honest reality check

## 5.1 Joint re-tuning overfits (V10)

Re-tuning all parameters jointly (score, blob threshold, $\tau_2$, X3D gate, morphology) found a validation optimum (val F1 55.4%) that *dropped to 54.4% on test* — below config-D's 60.8%. With only 40 validation positives, tuning ∼6 knobs fits noise.

## 5.2 Cross-validation: the true number

We ran 4-fold timeline cross-validation over the 1108-frame val+test pool (81 positives, every frame tested once, no detector leakage):

| Detector | fold F1s | mean ± std | pooled F1 |
|---|---|---|---|
| config-D rule | 32/51/65/0 % | 36.9% ± 24.3% | **44.3%** |
| logistic (no motion) | 34/58/42/0 % | 33.4% ± 21.1% | 42.6% |
| logistic (+ motion) | 29/52/36/0 % | 29.4% ± 19.0% | 37.8% |

**Findings.** (i) config-D's honest F1 is **≈44%**, not 60.8% — the single split was a favourable region. (ii) The fold-to-fold spread is enormous (±**24 points**, folds ranging 0–65%). (iii) **Motion features hurt** (−4 points) — frame-difference energy and merge run-length added noise the model overfit. (iv) Every learned variant again trailed the rule.

## 5.3 How much data would 65% actually need?

A bootstrap of config-D's pooled predictions gives 95% confidence intervals: frame-level [**35.7%, 52.4%**] (±8.3) and scene-level [**12.2%, 66.3%**] (±27). The scene-level interval — spanning almost the whole range because positives come from only four scenes — is the binding constraint. To estimate F1 to ±2.5% (so 65% is distinguishable from 60%) needs on the order of **∼900 positive frames (∼11× current)** and **3–4× more distinct contact scenes**.

## 5.4 Can old MLX90640 touch data substitute?

We pretrained Thermo-X3D on the older MLX90640 touch recordings (real contact, but 24×32, upsampled to 62×80) and fine-tuned on Waveshare. It **reduced** F1 by 6.2 points (scratch 35.9% → transfer 29.7%): the low resolution cannot reproduce the two-body structure the method relies on, and the sensor-domain gap caused over-firing. Synthetic compositing of touches was judged unviable (it must be geometrically consistent across all three cameras simultaneously).

# 6 Conclusions

1. **A practical homography-free detector exists.** "config-D" (MobileNet-SSD + residual blob-merge + two-body test + temporal smoothing) reaches F1 60.8% on a matched split and beats the deep Thermo-X3D baseline, while being explainable and needing no contact training labels.

2. **Transferable design lessons:** (a) the connected-warm-component *merge* test is the single most informative contact signal; (b) preprocessing should differ by task — *raw* frames for the detector, *residual* for blob segmentation; (c) temporal smoothing adds $\sim$9 F1 points; (d) homography self-calibration is too noisy to rely on here.

3. **Robust negative results:** every learned addition (logistic combiner, motion features, MLX transfer) and aggressive joint tuning matched or underperformed the simple rule — all symptoms of too few examples.

4. **The system is data-limited, not algorithm-limited.** Cross-validation gives F1 $\approx$44% with a [12%, 66%] confidence interval; 188 positives in four scenes cannot reliably train richer models or measure performance.

5. **Recommendation.** Adopt config-D as the working method and prioritise **collecting more contact recordings across more distinct scenarios** (target several times the current positive count and contact scenes). This is the single change expected to both raise and stabilise performance; further tuning on the present data is within measurement noise.